

Assignment 2: A Bayesian Test of Coin Fairness

CSDS 611 — Advanced Topics and Methods of Data Science I

Andrei Cozma

Data, hypotheses, and assumptions

The observed sequence is

$$D = (H, H, H, T, T, H), \quad n = 6, \quad k = 4 \text{ heads and } 2 \text{ tails.}$$

Let θ be the probability of heads on one flip. The hypotheses in the assignment are

$$H_0 : \theta = \frac{1}{2} \quad \text{and} \quad H_A : \theta \in [0, 1] \text{ is uncertain.}$$

I assume that, conditional on a common θ , the six flips are independent Bernoulli trials. To define the Bayesian alternative, I choose $\theta \mid H_A \sim \text{Beta}(1, 1)$, whose density is uniform on $[0, 1]$. For a decision, I assign $P(H_0) = P(H_A) = 1/2$ and choose the hypothesis with the larger posterior probability (the MAP rule). The assignment specifies none of these prior or decision choices, so the numerical decision below is conditional on them.

Likelihood and model evidence

For the particular ordered sequence D , conditional independence gives

$$p(D \mid \theta) = \theta^4(1 - \theta)^2.$$

Under the fixed fair-coin hypothesis,

$$p(D \mid H_0) = \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^2 = \frac{1}{64}.$$

Under H_A , the model evidence averages the likelihood over the stated parameter prior. Because the $\text{Beta}(1, 1)$ density equals 1 on $[0, 1]$,

$$\begin{aligned} p(D \mid H_A) &= \int_0^1 \theta^4(1 - \theta)^2 d\theta \\ &= \int_0^1 (\theta^4 - 2\theta^5 + \theta^6) d\theta \\ &= \frac{1}{5} - \frac{1}{3} + \frac{1}{7} = \frac{1}{105}. \end{aligned}$$

These two evidences describe the same ordered data. If the data are instead represented only by the count of four heads, each evidence is multiplied by $\binom{6}{4} = 15$, leaving their ratio unchanged.

Comparison and decision

The Bayes factor, oriented as alternative versus null, is

$$\text{BF}_{A0} = \frac{p(D \mid H_A)}{p(D \mid H_0)} = \frac{1/105}{1/64} = \frac{64}{105} \approx 0.610.$$

Thus these data favor H_0 over H_A by a factor of $105/64 \approx 1.64$. With equal prior probabilities for the hypotheses, Bayes' rule gives

$$\begin{aligned} P(H_0 \mid D) &= \frac{(1/64)(1/2)}{(1/64)(1/2) + (1/105)(1/2)} = \frac{105}{169} \approx 0.621, \\ P(H_A \mid D) &= \frac{64}{169} \approx 0.379. \end{aligned}$$

The MAP rule therefore selects H_0 : under the stated assumptions, the six flips provide modest evidence in favor of the fair-coin model. This decision does not establish that the coin is fair with certainty.

The prior probabilities matter. For general prior odds $r = P(H_A)/P(H_0)$, the posterior odds are

$$\frac{P(H_A \mid D)}{P(H_0 \mid D)} = \frac{64}{105} r.$$

The MAP rule would instead select H_A if $r > 105/64$. Consequently, the assignment's hypotheses and six observations alone do not determine a unique Bayesian decision without a prior and decision rule. Although the stated interval for H_A includes $\theta = 1/2$, a continuous uniform prior assigns zero probability to that single value within H_A ; H_0 remains a distinct model concentrated at $\theta = 1/2$.